Sri Lanka Institute of Information Technology



Project Topic: Web-based Boat Safari trip management.

IT24100889	Henarangoda M.P.
IT24100824	Gunasena P.G.P.
IT24100749	Hansaja H.A.
IT24100734	Lakshan M.S.
IT24100816	Jayasena W.M.S.A.
IT24510019	Wickramarachchi W A D M

Software Engineering | SE2030

Submission Date: 2025 / 10 / 26

2. Introduction

Safari Boat Trip Management System is a web-based application that would help to streamline and improve the organization and management of safari boat tours. As eco-tourism and nature-themed travels are becoming ever more popular, especially in such beautiful places as Sri Lanka, there grows a necessity in such reliable digital solutions that could help travelers to plan their trips and monitor them hassle-free. This platform supports both tourists and service providers since it offers a centralized and user-friendly interface through which one can book, monitor and manage safari boat adventures. It allows travelers to browse tours, check on-demand bookings, book and get confirmations and notifications. Meanwhile, operators and administrators are able to regulate trip schedules, assign staff, track ridership and receive inquiries by customer via a separate backend interface. The system also can enhance important features of operation as online payments, staff assignments, automated notifications, and trip status updates, to organize an excellent and professional experience of all users. The platform minimizes human error, time-consumption as it automates a large part of manual tasks typically related to trip coordination and generally makes the process more efficient. Also, the platform encourages sustainability through the reduction of paper-based reservations and enhancing more efficient management of limited eco-tourism resources, such as available boats, and employee assignments. The Safari Boat Trip Management System, with its focus on accessibility and user-friendliness and reliability, will serve the demands of present-day travelers and of eco-tourism enterprises wishing to digitalize their services.

The objectives and goals

The main aims of this system are as follows:

To make online booking and cancellations so that customers may get convenience and the staff may reduce work pressure.

Show real time availabilities on the boat trips to avoid over bookings and to be updated with the same.

The design of an administration panel that will enable creation of trips, assigning of user roles and reporting.

Automate Notification and Confirmations, the likes of email or SMS messages that are sent to the user when they successfully make a booking, a reminder or cancel trip.

Target Users and Stakeholders

Tourists- they will use the system to get knowledge on trip, make reservations and get updates on the trip.

Admin Staff- it will be in charge of the backend processes such as scheduling and coordination of trips and handling of data.

Boat Operators- those who will access their scheduled services and see to it that they are able to operate on time.

Tour Coordinators – monitoring the organization of the trip in general and interaction of the stakeholders.

Scope and Limitations

Scope

Web-Based Boat Safari Trip Management System is designed to facilitate easy and automated management and organization of boat safari trips.
Our system provides tourists with the ability to register, log in, view available safari trips, make online bookings, and tourists can provide a review of their experience.
Administrators can plan safaris, manage boat information, trip advertisements, and customer history using a secure web interface.
The system also allows marketing coordinators to control promotional activity and maintain trips feedback in their currency.
It is fully web-based and accessible via any standard web browser.
The main aim is to improve operational efficiency, reduce manual record-keeping, and overall customer satisfaction.

Limitations

Despite the system offering effective online booking and management functionality, it has a number of limitations.
It lacks live GPS tracking of ships or automatic weather condition updates.
The system depends on a constant internet connection and is not optimized to be used offline.
Online payment integration might be limited to specific payment gateways.
Refund management, superior analytics, and mobile app support are not included in the current version.
Future modifications can fill the gaps by implementing additional features and making it more accessible.

3. **Requirements**

Functional Requirements

1. User Registration & Login

The system will allow users, including tourists, staff and administrators, to register and securely log in to their accounts. There will be role-based dashboards that will guarantee that every user only has access to features that is relevant to their responsibilities. Email verification, password retrieval, and profile management are some of the features that will facilitate a secure and reliable login experience.

2. Trip Browsing & Search
Tourists will be able to browse available safari trips and find them by date, time, boat availability, and price. Detailed information such as boat type, guide details, and seat availability will be displayed, making it easier for customers to select a suitable trip.

3. Online Booking & Payment
The system will enable tourists to reserve seats and confirm bookings over the internet. Secure payment gateways will be integrated for payment via receipt of bank slip. Upon successful booking, customers will be confirmed.

4. Trip & Resource Management
Administrators and booking managers will manage trip details through the system. This includes creating, editing, and deleting trips, assigning boats and guides, setting capacities, and adjusting prices. The feature ensures resources are allocated efficiently and no scheduling conflicts occur.

5. Booking History Viewer
The customer will be able to view their booking history on their accounts. They can view upcoming trips booked, past bookings, payment history, and download receipts. Cancel or modify booking options will also be available, subject to the company's policies.

6. Trip Schedule Dashboard
The operators of boats and tour guides will have access to see their allocated schedules in a designated dashboard. This will consist of dates, hours, schedules and the list of passengers and any changes to cancelations or delays. The aspect will enable employees to be updated and make fewer mistakes.

7. Promotions & Reviews Management
The marketing personnel will have the opportunity to design and change the available offers, discounts, and highlights of the trips. The system will also enable reviews by customers to be posted so that the customers can read them and become more engaged and trust the system. This would be a valuable feature because it will attract new tourists, and it will help this company to create its virtual presence online.

8. Notifications & Alerts
The marketing personnel would be capable of creating and updating promotional offers, discounts, and highlights of trips. The system will also enable customers to write reviews and display them thereby increasing customer trust and interaction. The feature can be useful in getting new tourists as well as improve online presence of the company.

Non-Functional Requirements

Performance

The system must be efficient and able to cater to the number of users simultaneously without causing delays. It must at least support 100 simultaneous users so that tourists, staff, and administrators will also feel free to navigate the site even at times when there is peak traffic.

Security

Security is a critical requirement given that personal information contained in the system includes the user logins and reservations information. There will be secure authentication on the user information and secure storage of sensitive information in-between.

Usability

The site ought to be easy to navigate around to ensure that the tourists and the staff do not waste a lot of time as they progressively search the site and find what they are interested in. The system must also be adaptive on various devices such as the smartphone, and tablet.

Reliability

Availability of the system means that it works every time it should work so that the users can use it. No outages should be allowed and there should be frequent backup of data to prevent loss of data. In case of systems failure, the system should have a simple recovery strategy so that there is quick restoration of programs.

Scalability

The system must be scalable i.e. it must be able to increase with the growth of the business. In future, RiverTrack Safaris might introduce additional number of boats, guides or even launch new safari destinations; such additions should be acceptable depending on the system used and it should not require a big change.

Maintainability

The site must be very simple to keep and update. The personnel must have the possibility to make adjustments to trip information, promotions or schedules without having a high amount of technical knowhow. This will make sure that the system is current and applicable in the day-to-day operations.

System Limitations / Constraints

1. Web-Only Access:

The system is currently limited to a web-based platform, with no native mobile application. This restricts users who prefer to manage bookings via mobile apps and limits access in areas with poor browser support.

2. Internet Dependency:

All features require an active internet connection. Users in remote areas with weak internet signals may face difficulties using the system, especially during payment or confirmation steps.

3. Limited Payment Options:

Initially, the system supports only card-based online payments. It does not support alternative methods like bank transfers, mobile money, or cash-on-arrival, which may be preferred by local users**.**

4. Single Provider Focus:

The initial version of the system is designed to serve a single safari service provider. This limits the platform's ability to serve multiple competing businesses or operate across multiple regions.

5. Lack of Real-Time GPS Tracking:

There is no integration with GPS or live tracking for boats. Users and administrators cannot monitor the real-time location of tours, which could improve safety and customer experience.

6. No Weather Data Integration:

Weather plays a significant role in safari planning. However, the system lacks integration with real-time weather services, which could otherwise help in rescheduling trips and improving safety.

7. Basic Admin Controls:

Admin functionality in the first release includes basic CRUD operations only (Create, Read, Update, Delete). It lacks features like analytics, advanced reporting, user activity logs, and detailed permission management**.**

8. Limited Customer Support Features:

The system initially provides only email support and FAQ sections. It does not include live chat, chatbot integration, or a helpdesk dashboard, which can delay user issue resolution**.**

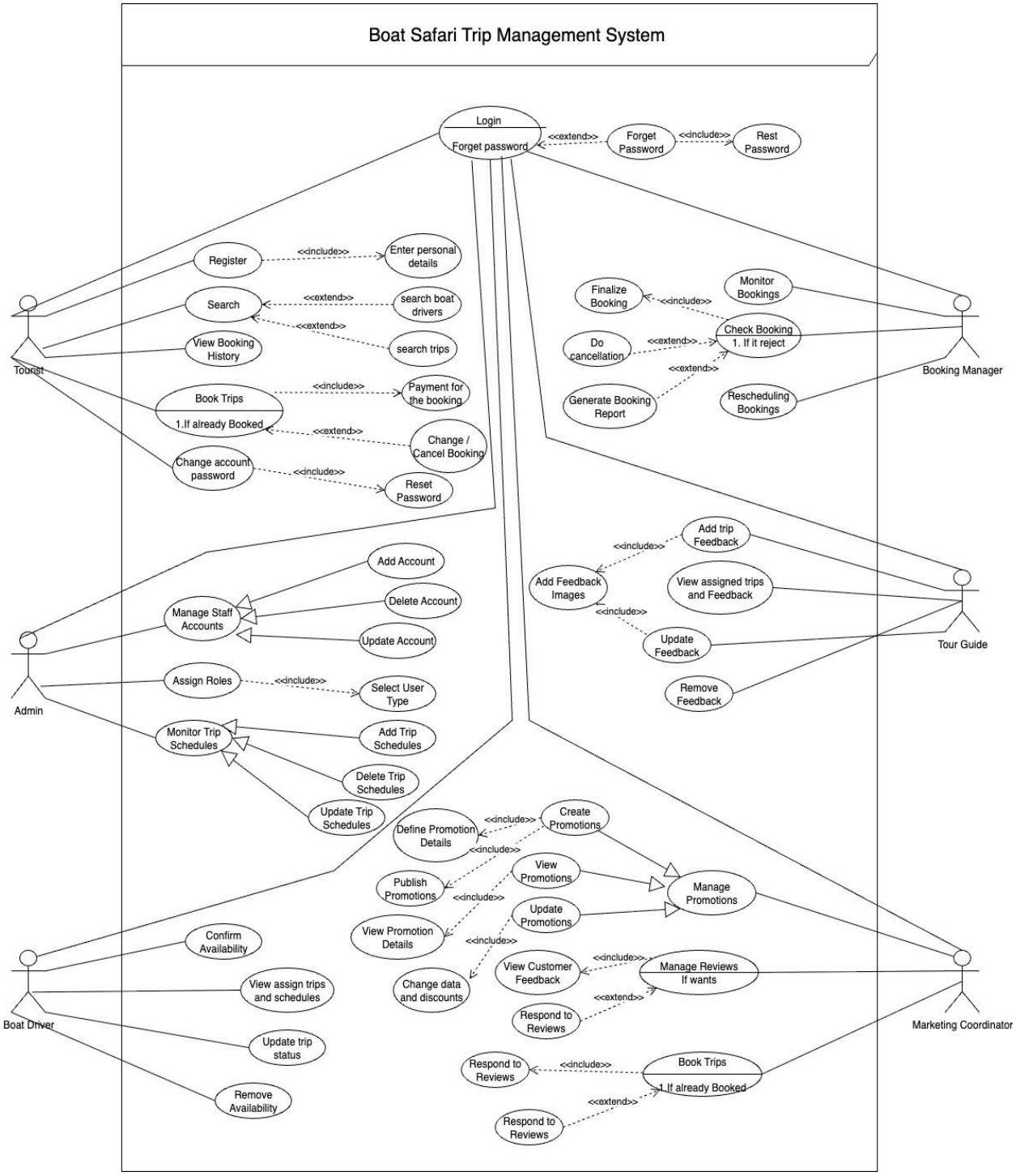
9. Scalability Limitations:

While designed to support future expansion, the first version may struggle under high traffic or in multi-location environments due to limited server infrastructure and codebase optimizations.

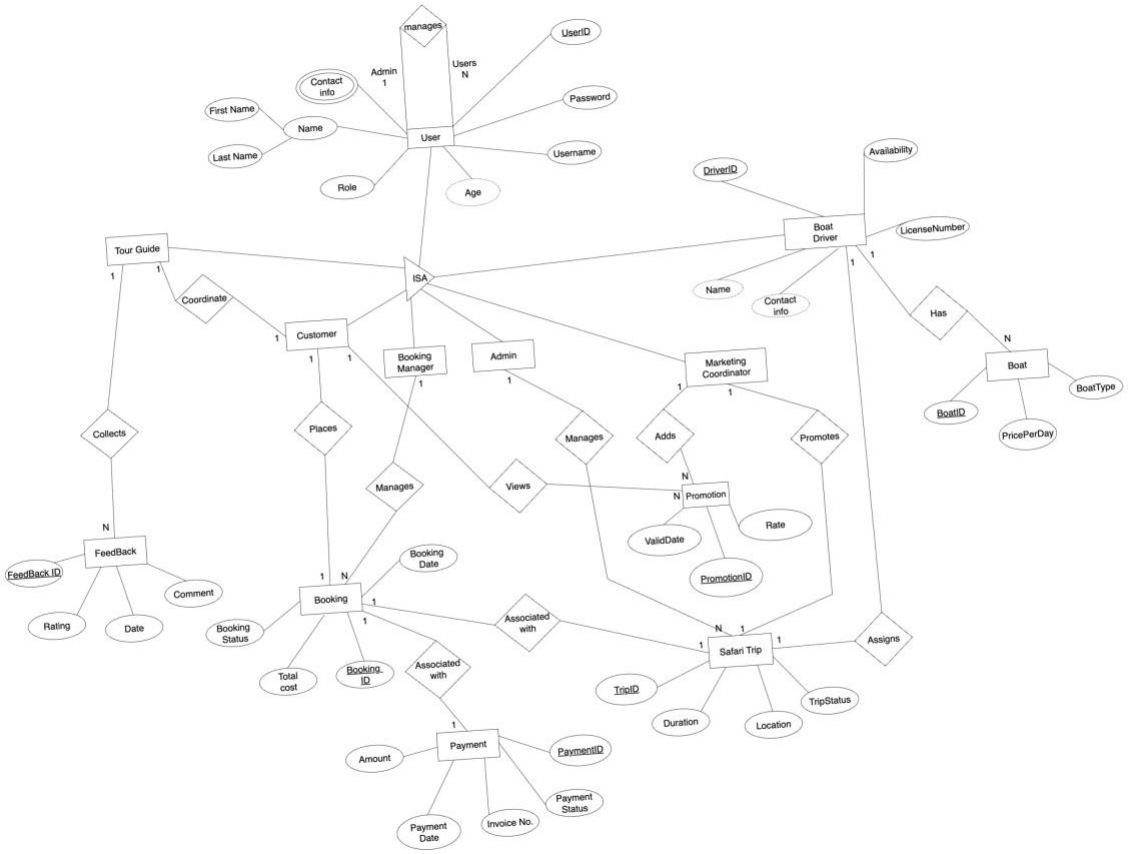
10. Language and Accessibility Constraints: The system is designed in English only and may not be fully accessible for non-English speakers or users with disabilities. Accessibility features like screen reader support or alternative text for images are limited.

4. Design

Use Case Diagram:

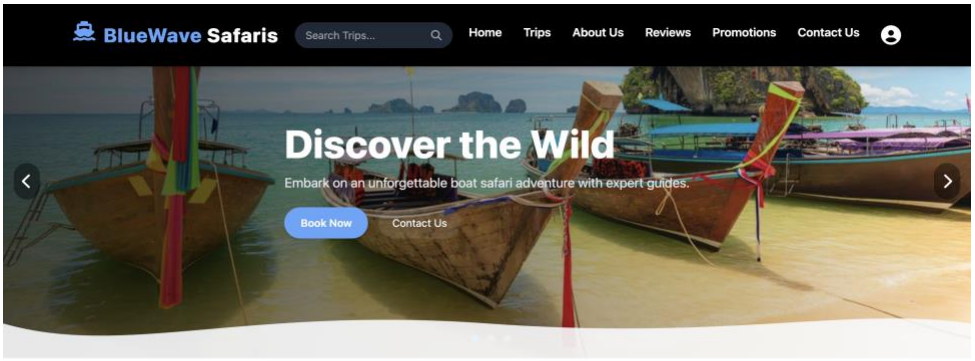

Boat Safari Trip Management System
Login
Forget password
<<extend>>
Forget Password
<<include>>
Rest Password
Register
<<include>>
Enter personal details
Search
<<extend>>
search boat drivers
<<extend>>
search trips
View Booking History
Tourist
Book Trips
1.If already Booked
<<include>>
Payment for the booking
<<extend>>
Change / Cancel Booking
Change account password
<<include>>
Reset Password
Finalize Booking
<<include>>
Monitor Bookings
Check Booking
1. If it reject
Do cancellation
<<extend>>
<<extend>>
Generate Booking Report
Rescheduling Bookings
Booking Manager
Add Account
Delete Account
Update Account
Manage Staff Accounts
Assign Roles
<<include>>
Select User Type
Admin
Monitor Trip Schedules
Add Trip Schedules
Delete Trip Schedules
Update Trip Schedules
Add trip Feedback
<<include>>
Add Feedback Images
View assigned trips and Feedback
<<include>>
Update Feedback
Remove Feedback
Tour Guide
Define Promotion Details
<<include>>
Create Promotions
<<include>>
Publish Promotions
<<include>>
View Promotion Details
<<include>>
Change data and discounts
View Promotions
Update Promotions
Manage Promotions
Confirm Availability
View assign trips and schedules
Update trip status
Remove Availability
Boat Driver
View Customer Feedback
<<include>>
Manage Reviews If wants
<<extend>>
Respond to Reviews
Respond to Reviews
<<include>>
Book Trips
1.If already Booked
<<extend>>
Respond to Reviews
Marketing Coordinator

ER Diagram:



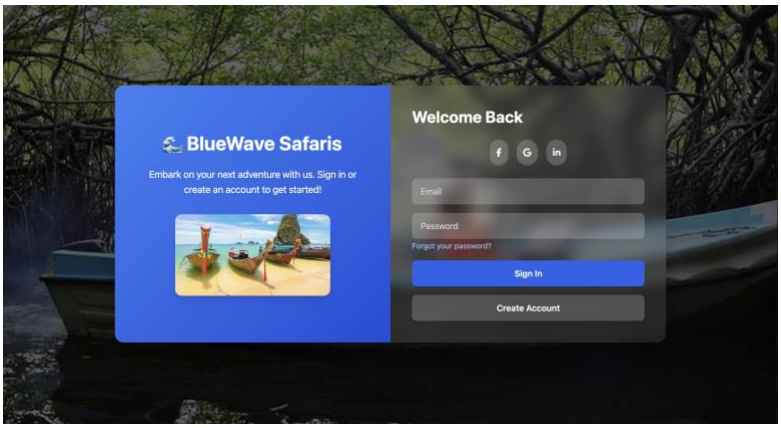
UI Sketches/Screenshots and Descriptions:

1. Home Page of the Web-Based Boat Safari Trip Management System.

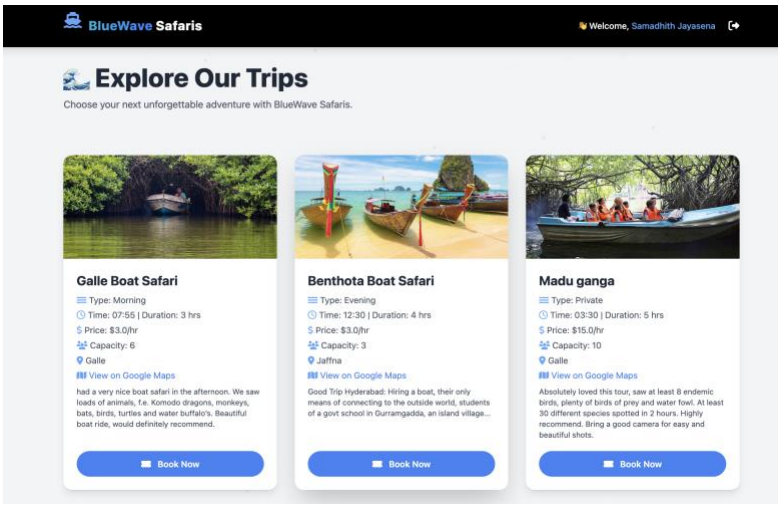




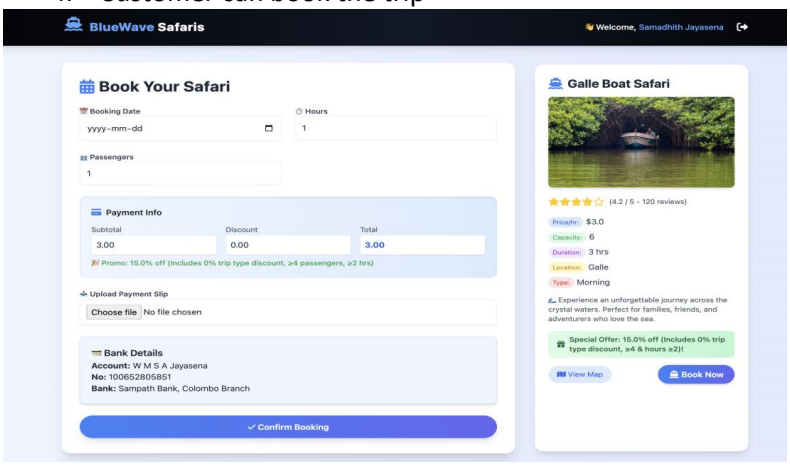
2. Login/Sign in Page of the system.



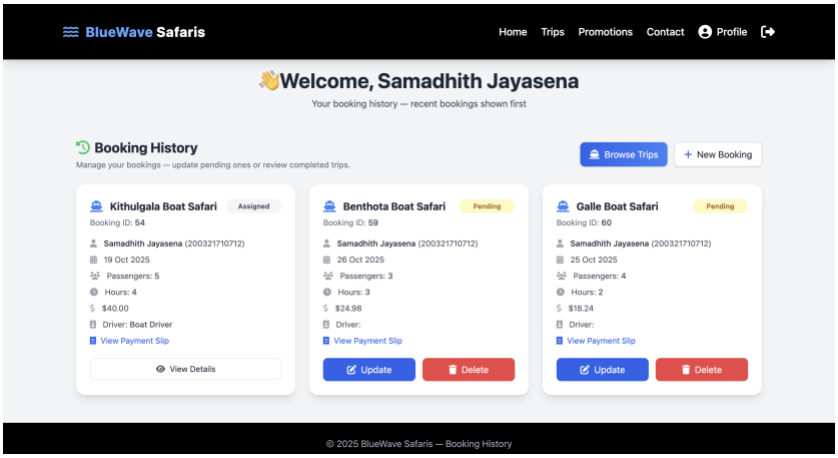
3. Admin assigned/created trips can view tourist.



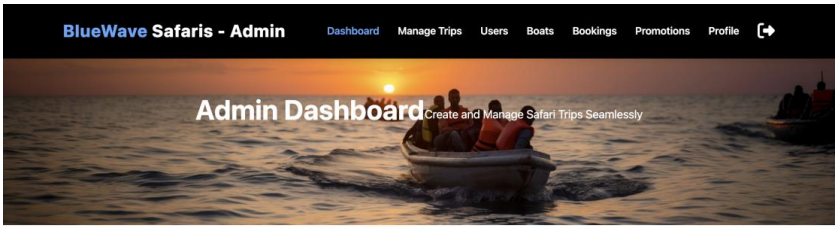
4. Customer can book the trip

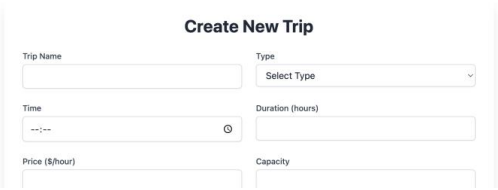


5. Tourist can change or cancel the booking.

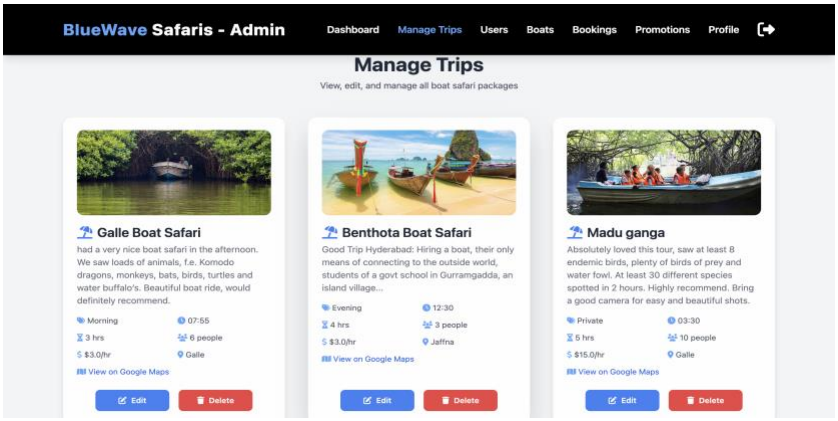


6. Admin can create trips

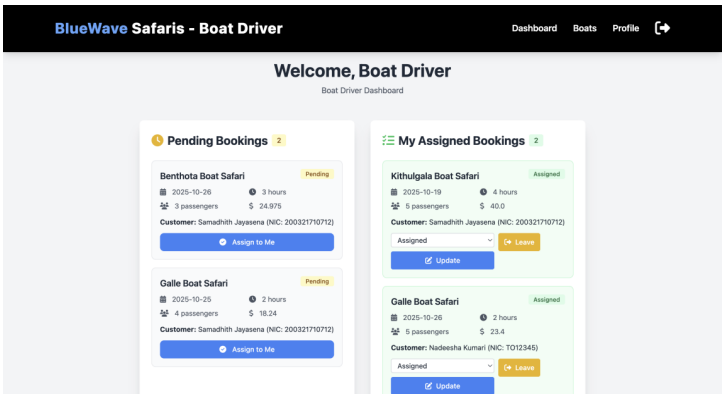




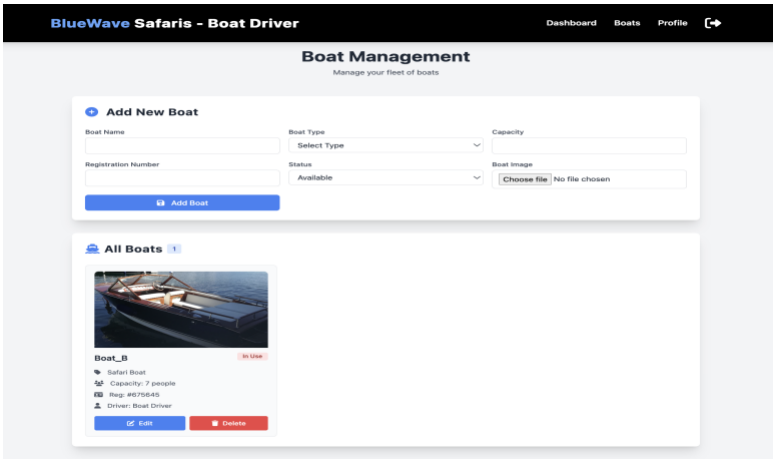
7. Admin can manage the trips.



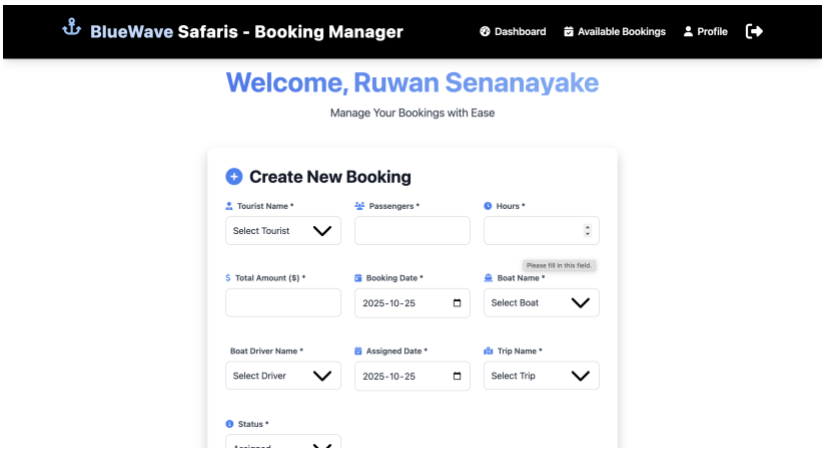
8. Boat driver can assign to a trip.



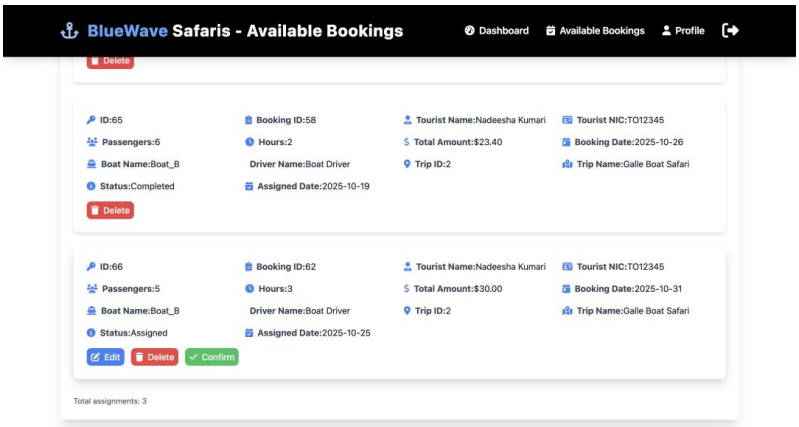
9. Boat driver can add boats.



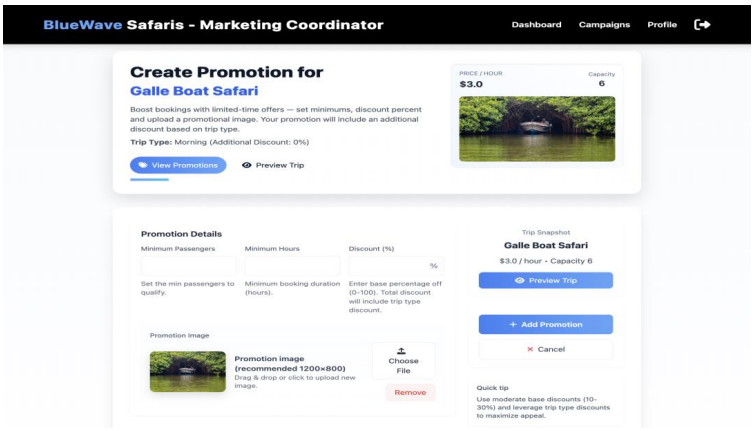
10. Booking Manager can add bookings manually.



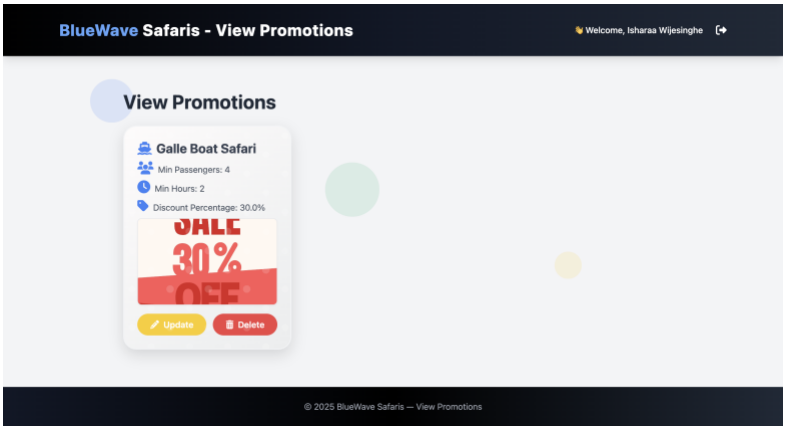
11. Booking manager can confirm, delete, update the tourist bookings.



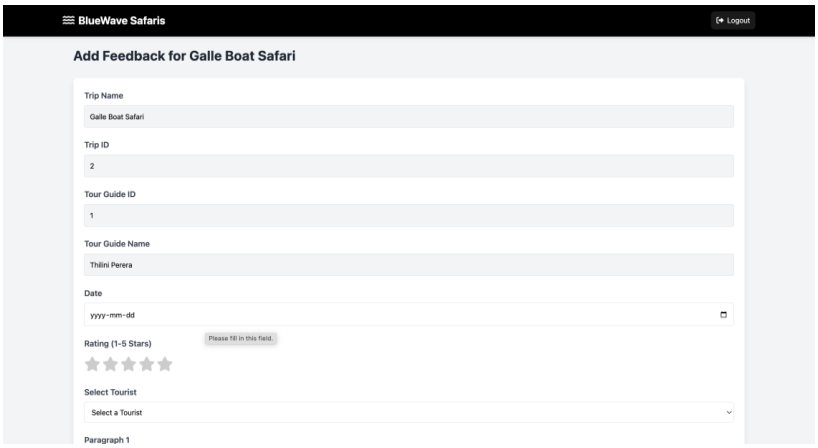
12. Marketing coordinator can add promotions/discounts for trips



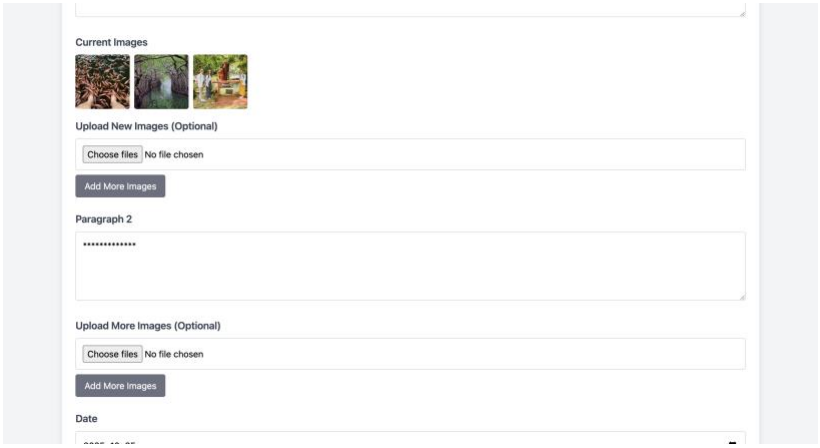
13. Marketing coordinator can manage the promotions.



14. Tour Guide can collect feedback and add to the system.



15. Tour Guide can Edit or Delete the added Feedback



Implementation

Tools and Technologies Used

- Programming Language: Java

- Framework: Spring Boot (uses @Controller, dependency injection with @Autowired, Spring MVC routing annotations like @GetMapping and @PostMapping)
- Design Patterns: Strategy Pattern (used for payment validation and promotion/discount calculation)

- IDE: IntelliJ IDEA Ultimate Edition

- Database: MySQL (via repositories like TripRepository, BookingRepository, PromotionRepository)

- Web Technologies: HTML, CSS, JavaScript
- File Upload Handling: MultipartFile for payment slips, saved to local uploads directory
- HTTP / Session Management: HttpSession for user login tracking
- MVC Architecture: Model-View-Controller pattern to separate concerns

Six Major Functions

1. **Trip Booking and payment (Customer/Tourist)**

Travelers these days are fond of easiness of use, and that is why the Trip Booking & Payment tool has been created with the given purpose in mind. The role of this function is that the tourists can navigate through the available safari boat tours when they are at the comfort of their home or using their mobile phones. They can compare trips, review descriptions, timings, lengths, and availability by traversing several clicks enabling them to make thoroughly informed decisions.

After selecting a favorite tour, the customers are able to ratify the reservation by having the booking through a secure and trusted online payment gateway. This is a virtual process which negates the conventional use of phone calls, physical visits or manual checks. In this way, the facility can save a lot of time through this and customer satisfaction is also enhanced due to the reasons that with this facility, the process of making bookings is not only becoming convenient but more time-saving as well. Finally, it provides simple, trustworthy, and convenient planning and booking experience that enables the customers to dedicate their attention to the anticipation of their new journey rather than concern about making accommodation reservations.

2. **Trip & Resource Management (Admin Officer)**

No great safari business would be complete without behind-the-scenes crew who had well-organized operations. The Trip & Resource Management section will be established to support the day-to-day logistical issues that are facilitated by the Admin Officer. This involves putting the right prices to all the safari package, making real time arrangements of boats and captains in tours, booking qualified tour guides and updating the calendar of trips.

This can also help to make efficient use of a smaller number of available resources like boats and manpower, without overlaps, bookings, or tardiness. Centralized control will make various wings of the business cooperate and ensures that the business can run like a clock. The Admin Officer is able to concentrate on operations along with improving the overall experience by doing more planning and thinking into the future.

3. **Trip Schedule Dashboard (Boat Captain/Driver)**

The boat driver module allows drivers to efficiently manage their assigned trips. Through the Trip Schedule Dashboard, drivers can view all upcoming trips for the day or week, check passenger lists, monitor arrival and departure times, and review any special notes related to the journey.

In addition, drivers have the ability to perform CRUD operations, such as adding, updating, or

removing boats assigned to specific trips, ensuring that each safari is properly managed. This functionality reduces the chances of miscommunication, improves operational efficiency.

4. Trip Info Access & Feedback (Tour Guide)

The safari guides are the face of the tour. This feature will assist them to deliver interactive and customized experience by providing the guides with trip information prior to the journey. This will consist of the needs of the passengers, special requirements, flight schedule and any other information that the customer notes at the time of booking.

The guides will also be able to gather feedback and report it at the end of every trip and establish a positive feedback loop that will enable the continuous improvement. The system helps mainly with providing the guides with the methods of getting to know their audience better and measure the success of every tour that allows improving the quality of the provided services and satisfying the customer. With this, the good trip will ultimately be a great trip.

5. Booking Management (Booking Manager)

This is one of the most vital roles of the Booking Manager as it ensures smooth running in the background. It provides a singular place of management to attend to all the back-end tasks regarding bookings. The possibilities of checking a new booking validated and cancellations or, possibly, last-minute changes are within the tool range, so the manager can easily monitor everything in a very clear, and up-to-date format.

It also consists of intelligent validation that assists in identification of possible errors such as book doubling or time overlap. It will provide consistency and correctness in data and eliminate the costly errors and keep the whole team in line. As such, both customers and workers are aware of their booking procedure.

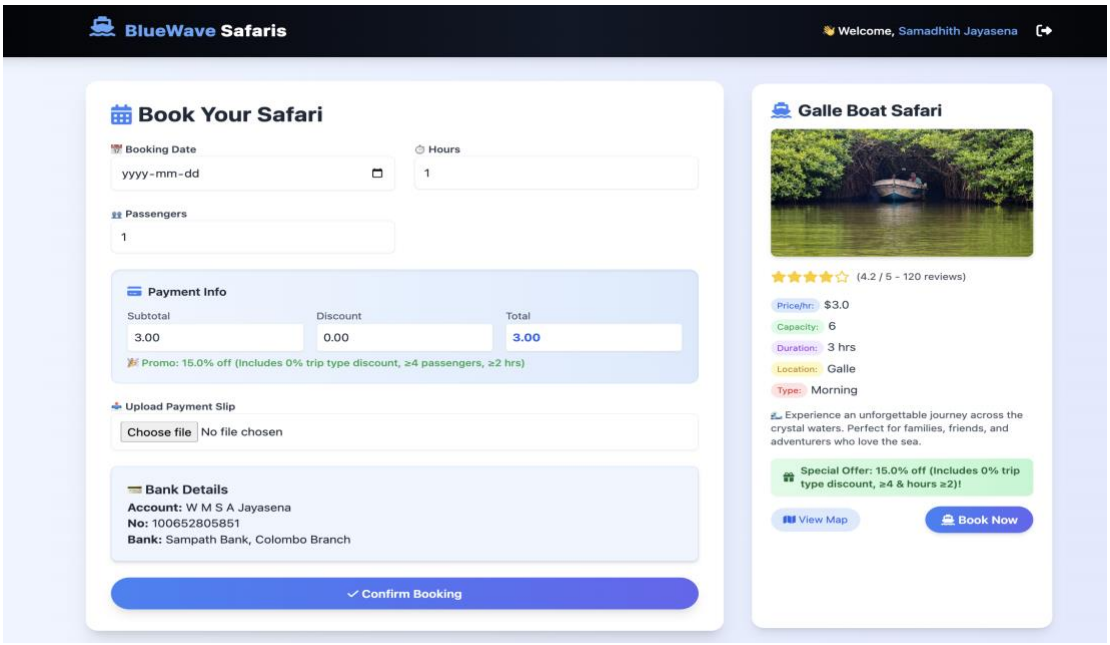
6. Promotions & Reviews Management (Marketing Coordinator)

Marketing is an important tool in the present computerized era, in attracting customers and retaining them. Using this tool, the Marketing Coordinator can create some appealing promotions, discounts, and season campaigns through the platform.

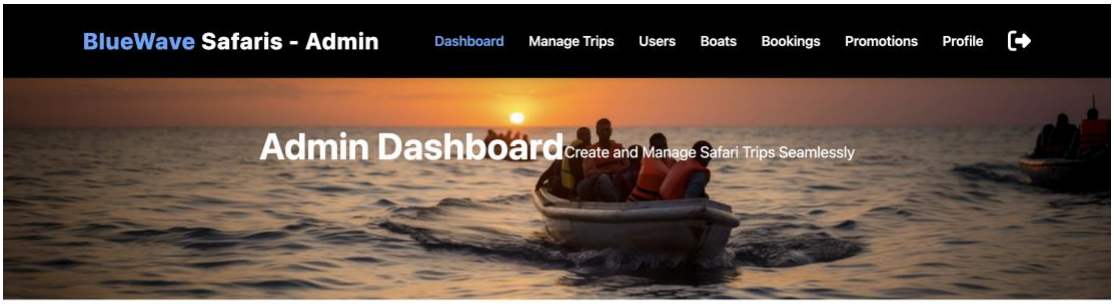
It offers an opportunity as well to obtain customer reviews and feedback in order to be able to post positive experiences at the social media or on the site. By combining the promotion and the real-time feedback, the software is more than just another generator of visibility and sales but also the garnering of trust into the business. Most people are ready to share their experience when they have seen real reviews and some inspiring stuff they have received.

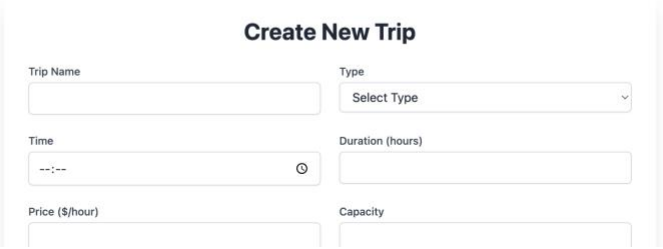
Screenshots of Core Functions

1. Trip Booking and payment

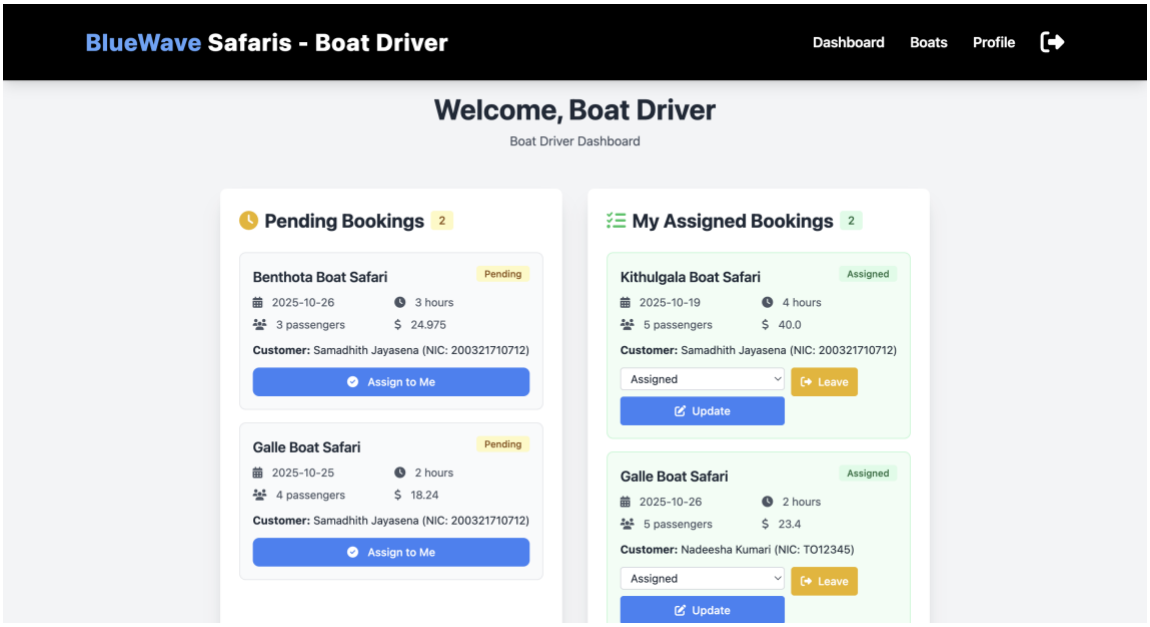


2. Trip & Resource Management

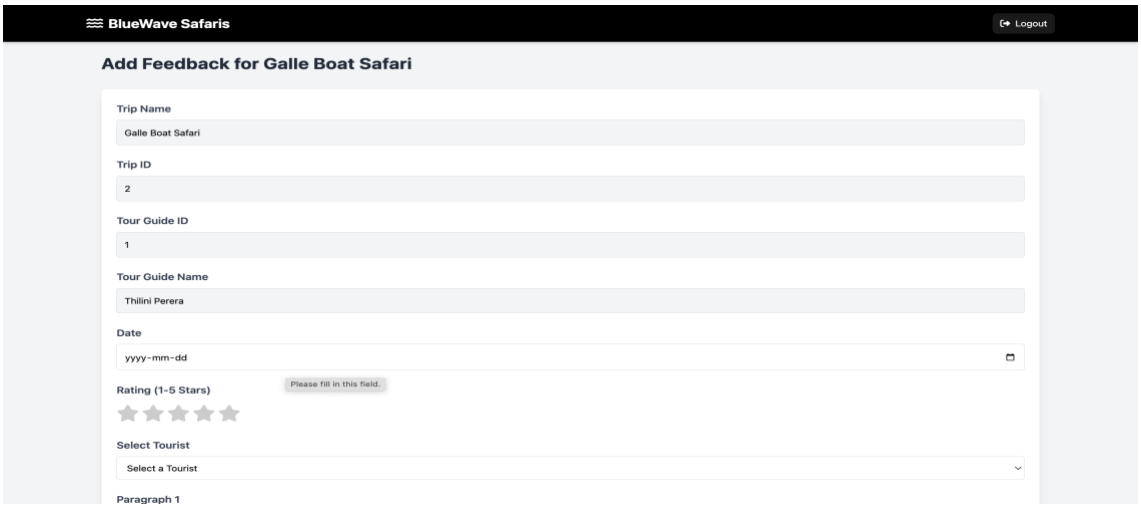




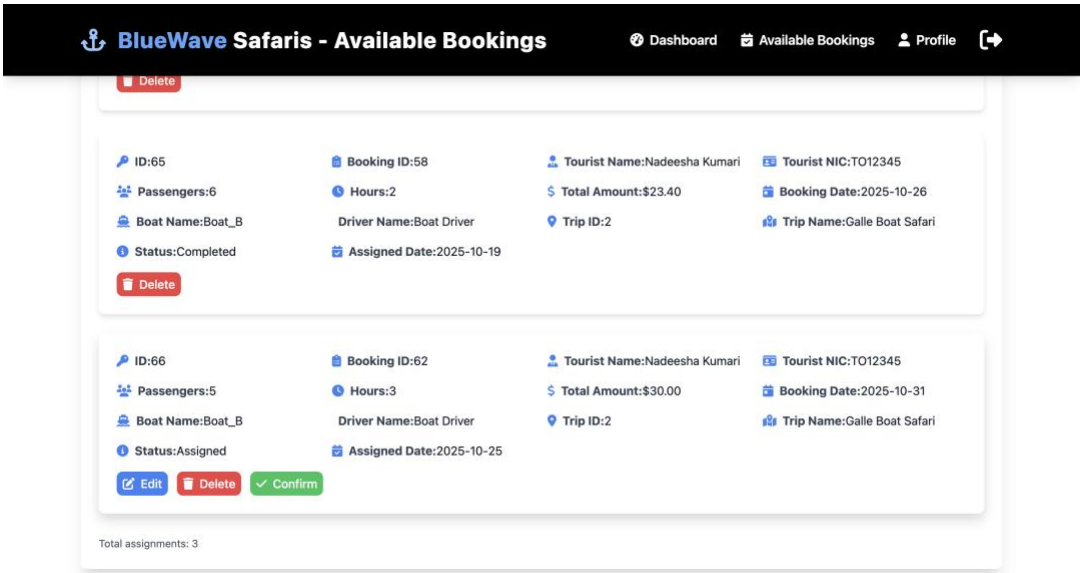
3. Trip Schedule Dashboard



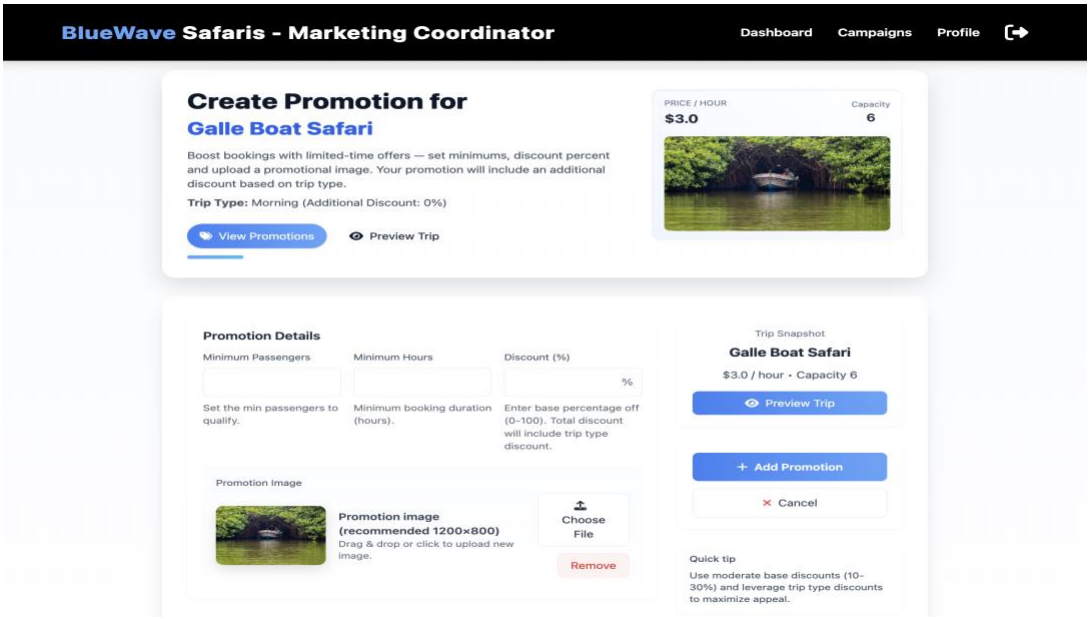
4. Trip Info Access & Feedback



5. Booking Management



6. Promotions & Reviews Management



6.Project Management

Task Distribution Among Team Members:

Sprint 1 – Core System Setup and Booking implementation

Task Allocation and Completion

Team Member	Responsibilities	Status
Hansaja H.A.	Developed booking module (search & booking flow)	Completed
Hansaja H.A.	Integrated secure payment gateway and validation	Completed
Lakshan M.S.	Implemented trip availability management and scheduling dashboard	Completed
Jayasena W.M.S.A.	Developed staff management and role assignment functionalities	Completed
Wickramarachchi W.A.D.M.	Implemented payment confirmation and admin report modules	Completed
Gunasena P.G.P.	Coordinated sprint tasks and handled testing/validation	Completed
Henarangoda M.P.	Defined requirements and validated sprint outcomes as Product Owner	Completed

Sprint 2 – Customer and Marketing Features Implementation

Task Allocation and Completion

Team Member	Responsibilities	Status
Hansaja H.A.	Developed booking history and cancellation/reschedule modules	Completed
Lakshan M.S.	View Registered users into the system	Completed
Jayasena W.M.S.A.	Added maintenance logging and refund management modules	Completed
Wickramarachchi W.A.D.M.	Implemented notification and review	Completed
	display modules	
Gunasena P.G.P.	Coordinated sprint testing and bug tracking	Completed
Henarangoda M.P.	Validated customer experience flow and sprint outcomes	Completed

Sprint 3 – Advanced Features and Final Integration

Task Allocation and Completion

Team Member	Responsibilities	Status

Hansaja H.A.	Developed multilingual trip content and promotions module	Completed
Lakshan M.S.	Integrated seasonal pricing updates and analytics dashboard	Completed
Jayasena W.M.S.A.	Handled final admin module refinements and testing	Completed
Wickramarachchi W.A.D.M.	Integrated social media sharing and system deployment	Completed
Gunasena P.G.P.	Coordinated integration testing and documentation	Completed
Henarangoda M.P.	Validated final system and approved release	Completed

7. Conclusion

Summary of Achievements:

We successfully developed our Web-Based Boat Safari Trip Management System to streamline and organize safari trip operations. HTML was used to design the front end, Java was used to build the back end, and MySQL was used for the database. The system has all of the primary functions we had in mind: boat drivers can assign boats for planned trips, customers can book trips, and the administrator can oversee all activities. While the booking manager manages trip reservations, tour guides have the ability to view and evaluate their trips. The marketing coordinator is able to handle consumer feedback and develop promotions. All things considered, we succeeded in creating a comprehensive and useful system that enhances coordination amongst all parties involved.

Challenges Faced:

Managing communication between various modules and seamlessly integrating the database

and back end were two of the challenges we encountered during development. Since a number of components needed to be finished and tested quickly, time management was another difficulty. However, we were able to resolve the majority of these problems and successfully finish the project by cooperating, testing often, and conducting more research.

Suggestions for Improvement or Extension:

To increase convenience and safety, we would like to add GPS-based boat tracking and an online payment option to the system if we had more time and funds. It would also be simpler for users to access the system at any time if an Android app or mobile-friendly version were developed. In the future, incorporating data analysis tools to examine trip performance and customer feedback may aid in improving the overall experience and business decisions.

8. Individual Contribution, Teamwork

Role	Challenges	Overcome from challenges	Key lessons learned	Reflection
Admin	Ensuring correct trip schedules and avoiding conflicts; handling database updates safely	Used validation checks, tested CRUD operations thoroughly, coordinated with Booking Manager for schedule consistency	Learned database management, role-based access control, importance of testing	Admin module worked well; could improve by adding automated schedule conflict detection
Customer	Understanding how to integrate booking with payment and promotion systems	Studied BookingController workflow, tested booking scenarios, asked teammates for database information	Learned user session handling, booking flow, feedback management	Customer experience is smooth; could improve by adding more user notification
Boat driver	Tracking boat status and availability for assigned trips	Maintained manual logs, coordinated with Admin and Booking Manager	Learned communication with team, responsibility management	Boats operated efficiently; could improve with real-time boat status tracking in system
Tour Guide	Coordinating trip assignments and communicating with booking manager	Used system-generated schedules, confirmed assignments manually	Learned schedule coordination, basic system usage, teamwork	Trip assignment worked; could improve by adding automated notifications to guides

Booking Manager	Designing correct promotion logic and integrating with booking system	Implemented file upload validation strategies, tested edge cases, logged calculations for debugging	Learned file upload handling, payment validation, Strategy Pattern application	Booking system works as intended; could improve by adding automated payment verification
Marketing Coordinator	Designing correct promotion logic and integrating with booking system	Used Strategy Pattern for discount calculation, tested different scenarios, coordinated with Booking Manager	Learned Strategy Pattern, promotion logic, teamwork	Promotions applied correctly; could improve by adding analytics for marketing campaigns

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10. Appendix

GitHub link: https://github.com/IT24100816/Boat-Safari-Trip-Management-System